S.M.Hasnat Fardin

CSE Student, BRAC University | Full-Stack Developer & ML Enthusiast

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About Me

I'm passionate about both full-stack development and machine learning. I've built web applications using the MERN stack and Next.js, as well as research-driven AI projects, including an interpretable EEG-based emotion recognition system for my thesis. I enjoy exploring diverse areas, from AI music generation to embedded systems and IoT, and I'm always enjoying the process of learning new tools and applying them to real problems. I'm looking to grow further as a developer and ML practitioner, and I love projects that let me combine both sides of my skill set.

Education

BRAC University, Dhaka 2022 - Present

Bachelor of Science in Computer Science and Engineering (CSE)
GPA: 3.82 (Tentative)

Thesis: An Interpretable KAN-Guided SNN Architecture with Channel-Wise Contribution Tracking for EEG-Based Emotion Recognition

Murari Chand College, Sylhet 2021

Higher Secondary Certificate (HSC) | GPA: 5.00

Cantonment Public School and College, Saidpur 2019

Secondary School Certificate (SSC) | GPA: 5.00

Technical Skills

Programming Languages	Python, Java, JavaScript, PHP, C, C++, SQL, 8086 Assembly
Frontend	React.js, Next.js, HTML5, CSS3, Tailwind CSS
Backend	Node.js, Express.js, REST APIs, PHP, JWT Authentication
Database	MongoDB, MySQL
AI / ML	PyTorch, Spiking Neural Networks (SNN), Kolmogorov-Arnold Networks (KAN), Deep Learning, Machine Learning, LSTM, Transformer, Variational Autoencoders (VAE), Reinforcement Learning from Human Feedback (RLHF), EEG Signal Processing
Tools & Platforms	Git, GitHub ,Prisma, Postman, Vercel, XAMPP, Arduino IDE, emu8086

Undergraduate Thesis

An Interpretable KAN-Guided SNN Architecture with Channel-Wise Contribution Tracking for EEG-Based Emotion Recognition (Team Project)

Python • PyTorch • Spiking Neural Networks (SNN) • Kolmogorov-Arnold Networks (KAN) • EEG • Deep Learning

- Designed and developed an interpretable EEG-based emotion recognition framework using a Spiking Neural Network (SNN) backbone and a Kolmogorov-Arnold Network (KAN) readout to classify four emotional states from the DER-VREEG EEG dataset.
- Implemented EEG preprocessing, spike encoding, temporal feature extraction, and a KAN-based channel-wise contribution analysis to improve the interpretability of emotion classification.
- Evaluated the proposed model using accuracy, F1-score, Cohen's Kappa, Matthews Correlation Coefficient (MCC), and confusion matrix analysis, demonstrating interpretable and biologically inspired emotion recognition.

Course Projects

Multi-Genre AI Music Generation using Deep Learning

Research Project | Python • PyTorch • LSTM • VAE • Transformers • RLHF

- Developed an AI framework for symbolic music generation using LSTM Autoencoders, Variational Autoencoders (VAE), Transformer Decoders, and RLHF.
- Trained and evaluated models on the Lakh MIDI Dataset using reconstruction loss, perplexity, and human evaluation.
- Improved generated music quality by 19.3% through reinforcement learning from human feedback (RLHF).

Farmer Connect (Team Project)

MERN Stack | React.js • Node.js • Express.js • MongoDB • JWT

- Developed the frontend and backend of the Wholesaler/Retailer Operations module in a MERN-based agricultural marketplace.
- Implemented produce search, Top Farmers, purchasing workflow, transport coordination, and order tracking.
- Designed and implemented RESTful APIs and integrated MongoDB for efficient data management and seamless frontend-backend communication.

Smart Service Mart (Team Project)

Next.js | JavaScript • Prisma • NoSQL • Tailwind CSS

- Developed the frontend and backend of assigned modules in a full-stack service marketplace.
- Implemented real-time booking and scheduling, automated email notifications using Nodemailer, complaint management, and a smart service provider recommendation system.
- Integrated APIs and database components to deliver a scalable booking platform.

Remotely Controlled Robotic Arm

Embedded Systems | ESP32 • Arduino • NRF24L01

- Developed the ESP32-based glove controller for gesture-controlled robotic arm operation.
- Implemented wireless communication using NRF24L01 and integrated flex, force, and temperature sensors.
- Performed system testing and debugging to improve responsiveness and reliability.

Dual Authorization-Based Dam Control

IoT | Arduino • I2C • Embedded C/C++

- Developed a smart dam control prototype using dual Arduino Uno microcontrollers communicating via the I2C protocol.
- Integrated water level, rainfall, ultrasonic sensors, servo motors, and water pumps for automated dam operation.
- Contributed to embedded programming, circuit implementation, and system debugging.

Clinic Management System (Team Project)

PHP • MySQL • HTML • CSS • JavaScript

- Developed a full-stack clinic management system for appointment scheduling, patient records, doctor management, and billing.
- Designed the relational MySQL database and implemented CRUD operations.
- Integrated appointment scheduling, room allocation, emergency services, and payment management.

Banking Management System (Academic Project)

8086 Assembly • emu8086

- Developed a menu-driven banking management system using Intel 8086 Assembly Language.
- Implemented secure account creation with password authentication, deposits, withdrawals, balance inquiry, and loan management.
- Implemented modular assembly procedures using registers, loops, conditional branching, and memory operations to develop core banking functionalities.

The Amazing Psychopath (Team Project)

Computer Graphics | Python • OpenGL • PyOpenGL • GLUT

- Developed a dynamic day–night sky system featuring animated clouds, stars, moon, and gradient sky transitions.
- Designed and implemented animated UI text, floating kill-streak indicators, and pixel-based visual effects for gameplay feedback.
- Created a blood trail rendering system with dynamic trail generation and fading effects to enhance visual realism after collisions.
- Collaborated with teammates to integrate visual effects into a complete 3D OpenGL game environment.

Awards & Certificates

- Awarded the merit-based scholarship by Brac University, Bangladesh.
- Supervised Machine Learning: Regression and Classification — DeepLearning.AI & Stanford University (Coursera)
- Web Development — BRAC University